UNIVERSITY OF CALGARY

DATA 602 PROJECT

Energy Consumption and Economic Growth

Exploring Energy-Use Patterns and Economic Development Across Countries

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Why Energy Matters for Economic Development

Energy conditions can shape productivity, costs and the way economies develop over time.

01 RELIABLE ENERGY

Supports daily activity, production and transportation. Unreliable supply can reduce productivity and increase business costs.

02 CURRENT CONCERNS

Energy supply, energy security and rising costs can extend into inflation, financial markets and wider economic conditions.

03 CHANGING PATTERNS

Energy use is not constant. Countries at different stages of development may follow different energy-consumption paths.

WHY UNDERSTANDING THIS MATTERS

Understanding energy-use patterns can support better long-term economic and energy decisions.

- **Energy & infrastructure**
  Plan for changing energy needs.
- **Economic planning**
  Understand energy–development patterns.
- **Investment decisions**
  Inform resource allocation.
- **Environmental policy**
  Consider growth, energy use and emissions.

These connections motivate a closer statistical look at energy use across countries.

Our Research Questions

We focus on differences in energy-use change and the relationship between energy use and GDP.

RQ1

GROUP COMPARISON

Did the change in energy consumption per capita from 2010 to 2021 differ between countries with lower and higher initial levels of economic development?

Welch t-test + permutation test

RQ2

ENERGY & GDP

What is the relationship between energy consumption per capita and GDP per capita across countries in 2021, and how much of the variation in GDP per capita is explained by differences in energy consumption per capita?

Simple linear regression

Data & Study Design

Global country-level data from the World Bank's World Development Indicators.

ANALYTICAL SAMPLE

142

countries with sufficient data
for both analyses

DATA SOURCE

World Bank World Development Indicators

KEY VARIABLES

Energy use per capita

kg of oil equivalent per capita

GDP per capita

PPP, constant 2021 international dollars

HOW THE DATA ARE USED

RQ1 • 2010 → 2021

Calculate 2010–2021 energy-use change. Split countries by median 2010 GDP per capita: lower vs. higher initial development.

RQ2 • 2021

Model the relationship between 2021 energy consumption per capita and 2021 GDP per capita.

Complete-case approach • no imputation for missing required analysis data

RQ1 — What Does the Data Show?

Energy-use changes differ sharply between the two initial-development groups.

HIGHER INITIAL DEVELOPMENT

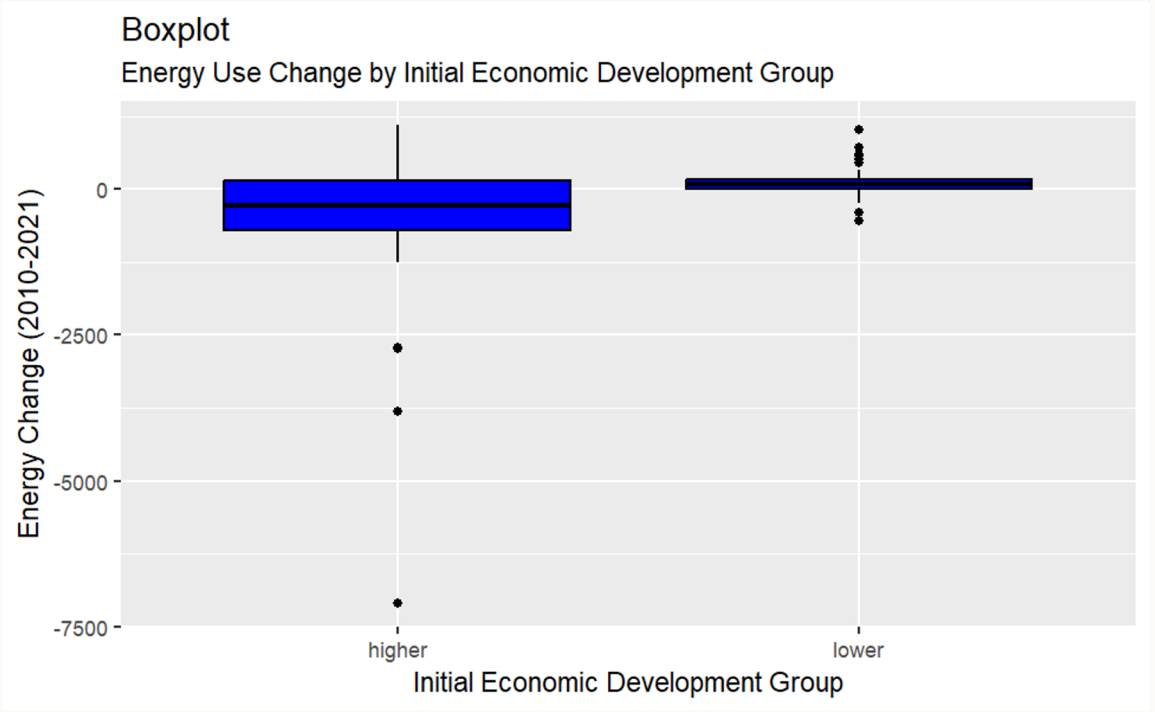
−445.8 kg SD = 1,115.3 kg

Mean change in energy use per capita, 2010–2021

LOWER INITIAL DEVELOPMENT

+121.7 kg SD = 230.6 kg

Mean change in energy use per capita, 2010–2021



Higher-development countries decreased on average, while lower-development countries increased.

RQ1 — Statistical Results

The two development groups show a statistically significant difference in mean energy-use change.

HYPOTHESES

Δ Energy = Energy Use$_{2021}$ − Energy Use$_{2010}$

H_0: $\mu\Delta,\text{Higher} = \mu\Delta,\text{Lower}$

H_a: $\mu\Delta,\text{Higher} \neq \mu\Delta,\text{Lower}$

STATISTICAL RESULTS

Welch t-test	t = −4.20 • p ≈ 0.00007
Permutation test	p ≈ 0.0005
95% CI (Higher - Lower)	[-836.7039 and -298.2768]

WHAT THIS MEANS

- There is sufficient evidence that mean energy-use change differs between the two initial-development groups.
- Higher-development countries decreased on average, while lower-development countries increased.

RQ1 — Sensitivity Analysis

Does the same pattern remain when the pandemic years are excluded?

We repeated the analysis using 2010 → 2019, while keeping the same development groups based on 2010 GDP per capita.



2010 → 2019 EVIDENCE

Welch t-test: p ≈ 0.0003 • Permutation test: p = 0.0005 • 95% CI: [−719.11, −227.42] kg

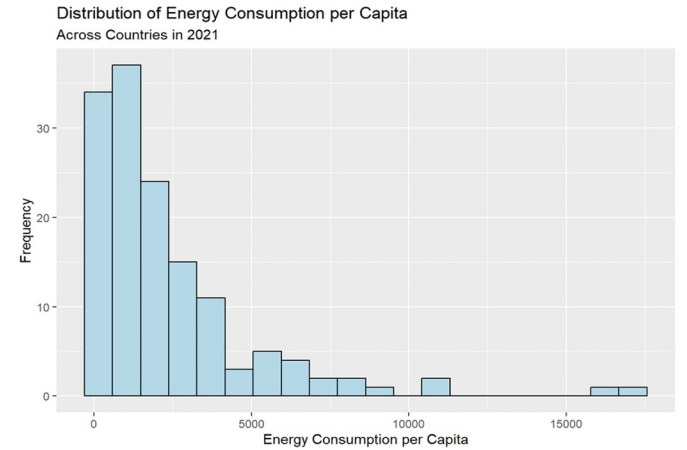
WHAT THIS MEANS

The gap becomes smaller, but the direction and statistical conclusion remain the same.
The difference in energy-use paths was already present before the 2020–2021 pandemic period.

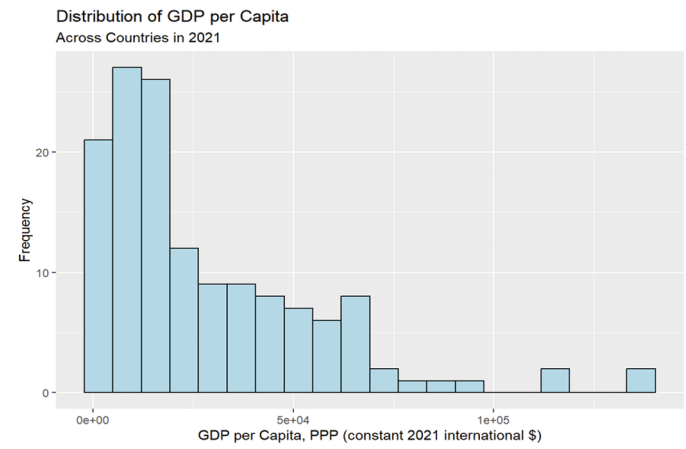
RQ2 — Exploring Energy & GDP in 2021

The distributions are strongly right-skewed, while the two variables show a positive association.

ENERGY USE PER CAPITA

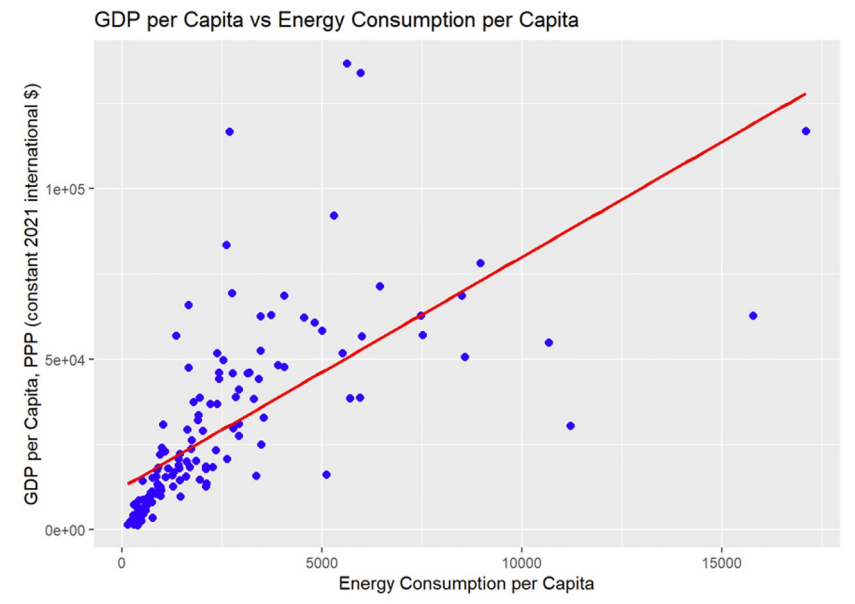


GDP PER CAPITA



The distributions of both variables are right-skewed

ENERGY USE vs. GDP



The scatterplot suggests a positive relationship between energy consumption per capita and GDP per capita.

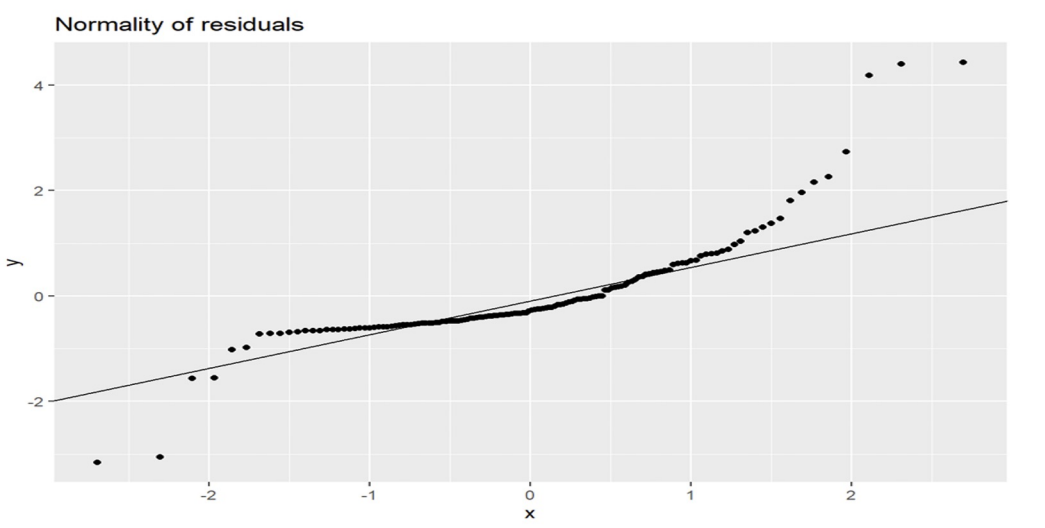
However, the extreme values and uneven spread suggest the original-scale regression conditions need to be checked.

RQ2 — Checking the Original Regression Model

The original-scale relationship is positive, but the residual diagnostics show that key model conditions are not satisfied.

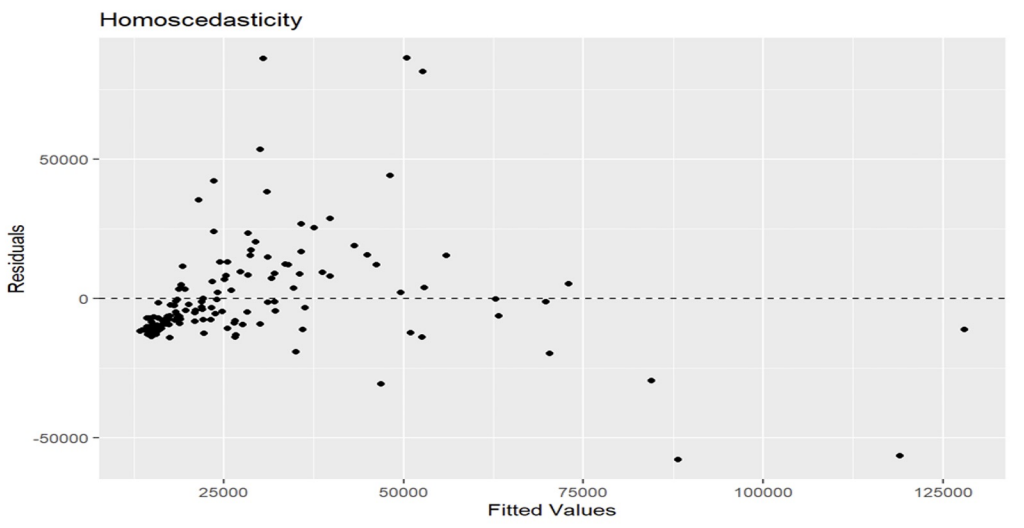
Original model: $\text{GDP per capita} = \beta_0 + \beta_1(\text{Energy consumption per capita}) + \varepsilon$

NORMALITY OF RESIDUALS



Residuals are not approximately normal.

HOMOSCEDASTICITY



Constant variance is violated.

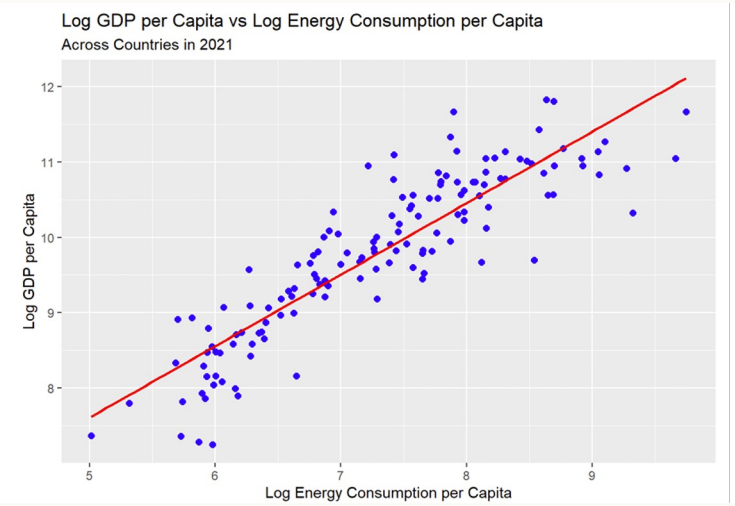
These diagnostics motivate transforming both energy use and GDP per capita before refitting the model.

RQ2 — Log Transformation Improves the Model

After the original model failed key diagnostic checks, both GDP per capita and energy use were transformed using logarithms.

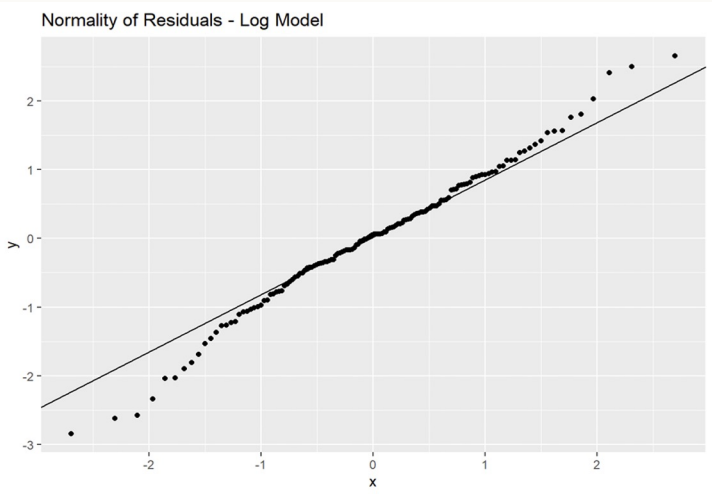
$$\log(GDP) = \beta_0 + \beta_1 \log(Energy) + \varepsilon$$

LOG–LOG RELATIONSHIP



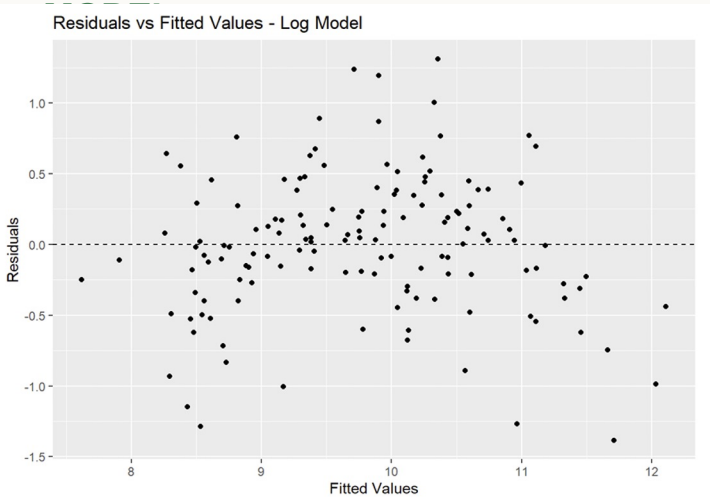
A clearer linear relationship after transformation.

NORMALITY — LOG MODEL



Residuals lie much closer to the reference line.

CONSTANT VARIANCE — LOG



Residual spread is more consistent across fitted values.

The regression assumption were reasonably satisfied, so the log-transformed model is used for the final RQ2 inference.

RQ2 — Statistical Results

How strongly is energy consumption per capita associated with GDP per capita across countries in 2021?

MODEL & HYPOTHESIS

$\log(GDP) = \beta_0 + \beta_1 \log(Energy) + \varepsilon$

$H_0: \beta_1 = 0$ $H_a: \beta_1 \neq 0$

ESTIMATED REGRESSION MODEL

$\widehat{\log(GDP)} = 2.8695 + 0.9479 \times \log(Energy)$

STATISTICAL RESULTS

Slope ($\hat{\beta}_1$)	0.9479
p-value	7.04×10^{-49}
95% CI	[0.8656, 1.0303]
Pearson r	0.8872
R^2	0.7871

WHAT THIS MEANS

A 1% increase in energy use per capita is associated with about a 0.95% increase in GDP per capita, on average.

The p-value is far below 0.05 and the 95% confidence interval does not include 0.

STRENGTH OF THE RELATIONSHIP

r = 0.8872 Strong positive linear association

R^2 = 0.7871 78.71% of the variation in log GDP per capita is explained by log energy use.

Final Discussion

The two analyses tell a more broader story than simply “more energy means more economic development.

RQ1

Energy-use trajectories differ across development stages.

- From 2010–2021, lower-development countries increased energy use per capita on average, while higher-development countries decreased.
- The pattern was not solely a pandemic effect.

RQ2

Across countries, higher energy use is closely associated with higher GDP per capita.

- A 1% increase in energy use was associated with about a 0.95% increase in GDP per capita.
- The model explained 78.71% of variation in log GDP.

Taken together, the results suggest that energy and economic development are strongly connected, but the relationship is dynamic: countries with higher income tend to use more energy, while changes in energy use over time may follow different paths as economies develop.

These findings describe cross-country patterns and associations; they should not be interpreted as evidence that changes in energy use cause changes in GDP.

Thank You for Listening

Questions & Discussion